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OS-Tutor-RAG

A retrieval-augmented Socratic OS tutor, and a study of how the choice of gold reference changes benchmark conclusions

CC: BCSE307P CN: Operating Systems

Adhyan Jain (24BCI0111) · Atharva Sheersh Pandey (24BCI0069)

Faculty: Dr. Jagalingam · October 2026

Introduction & Objective

Why a grounded chatbot is not yet a tutor

The problem

- **Slides are passive.** Terse decks and dense chapters never explain; they assume you know what to look for.
- **General chatbots are ungrounded.** Fluent, but a student cannot tell an explanation from a plausible invention.
- **Plain RAG is still a search box.** It answers what was asked and gives a beginner no way in.
- **Benchmarks hide differences.** One stored gold answer per task is arbitrary when many answers are correct.

Our idea

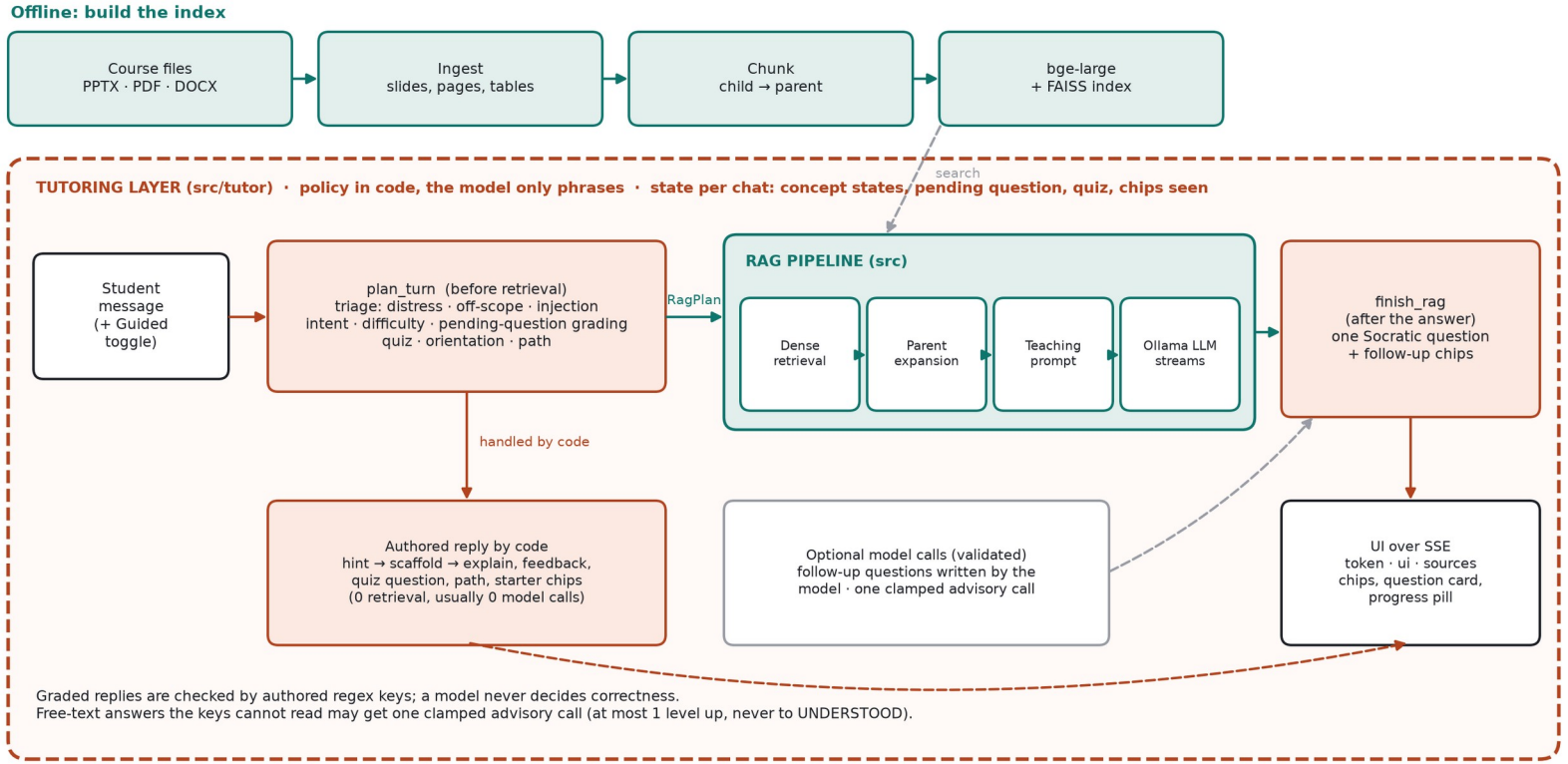
Ground every answer in the course files and let tested code decide how to teach; measure the evaluation, including its reference choice.

Objectives

- **O1** Index PPTX, PDF and DOCX without silently losing content.
- **O2** Compare chunking, retrieval, reranking and prompts on a 26-question golden set; adopt only what measures better.
- **O3** Measure the evaluation's own noise so small differences are not over-read.
- **O4** Streaming chat: ELI5 / undergrad / exam-prep levels, cited sources.
- **O5** Socratic tutor: 30 concepts, 96 graded questions, hint → scaffold → explain, 16 misconception probes, quiz mode, kind handling of struggle and distress.
- **O6** Orientation, learning path, starter and follow-up chips for a student who knows nothing.
- **O7** Research: measure how much scores, rankings and significance change when only the gold reference is swapped for another valid answer (RCR), and test it on stronger models.

Methodology

The tutoring layer wraps the RAG pipeline on every turn



Chatbot: small-to-big retrieval (match small chunks, read the parent slide) · policy in code (state, grading, next question) · the model only phrases

Research protocol (RCR): 24 executable OS tasks (scheduling, synchronisation, Banker's) · every valid answer V(x) enumerated · three evaluators: E1 exact match, E2 normalised match against ONE reference, E3 validator that replays the answer and never sees a reference · 50,000 sampled reference sets · 4 local models + 4 Claude models (1,152 generations each) · world-level sign-flip test, Holm correction

Results & Discussion (1/2): The chatbot, prompt and RAG findings

Measurement overturned the plan: fixing defects and the prompt beat choosing techniques

Scored prompt ("few_shot")

```
Answer the question using only the context
below. If the context doesn't contain the
answer, say so.

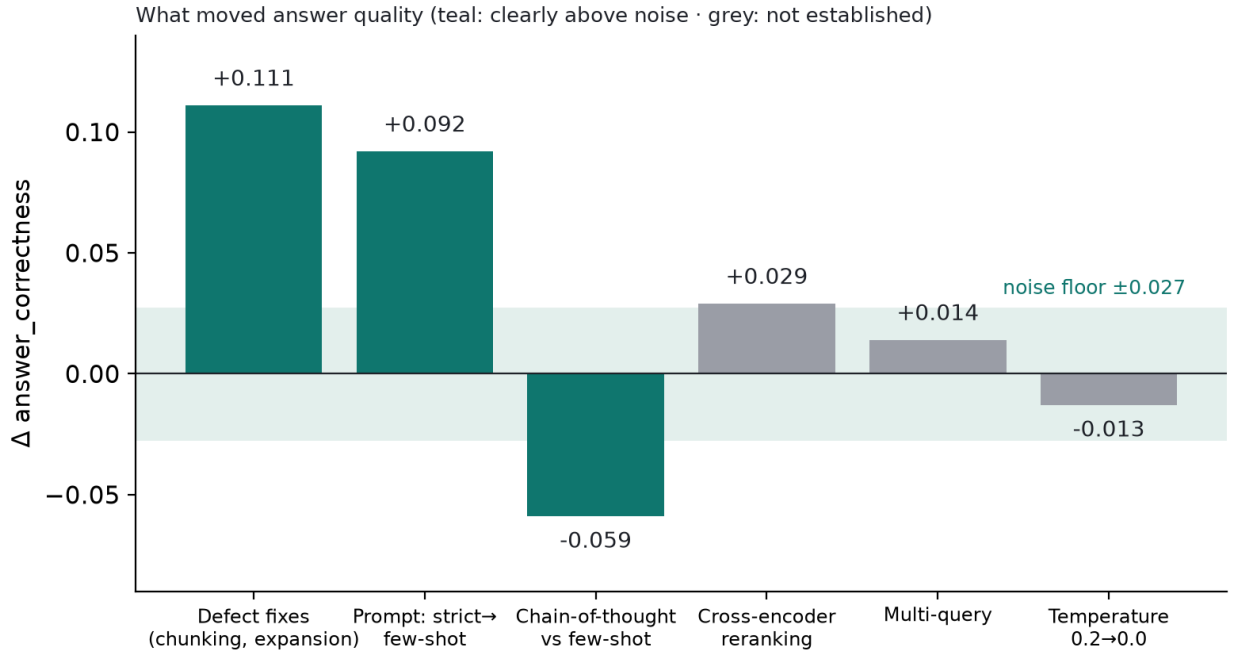
Answer in continuous prose, stating each fact
once. Do not add preamble about the context.

Example question: What are user mode and
kernel mode?
Example answer: CPU hardware has multiple
privilege levels: user mode runs user code,
and kernel mode runs OS code ...

Context: {context}  Question: {query}
```

Scoring: RAGAS, judge qwen2.5:7b (a different model family from the llama3 generator). Per question the judge sees question, retrieved contexts, answer and ground truth; answer_correctness is an F1 over statements. 26 questions, 5 topics.

The app's teaching prompt (explain first; the tutor appends the question) is a different prompt and was not RAGAS-scored.



Defects

Titles indexed as chunks: refusals 11.7% → ~0% once fixed.

Prompt shape

+0.092 few_shot; CoT −0.059 (answers 27% shorter).

Noise ±0.027

Same config scored 0.7752 and 0.7478. Dense R@5 = 1.000.

Results & Discussion (2/2): Research track (RCR)

Matching one stored answer hides real differences; a validator does not

The idea in one task

```
sched_01: FCFS, P1=24, P2=3, P3=3.
Six orderings are all valid:
  P1,P2,P3 · P1,P3,P2 · P2,P1,P3
  P2,P3,P1 · P3,P1,P2 · P3,P2,P1
Stored gold = #1. A model giving #4 is
equally correct but marked wrong.
E2 passes a valid answer with probability 1/|V|.
E[E2] = E3 / |V(x)|  (r = 1.000)
```

Follow-up: four Claude models above the floor (preregistered)

Model	Valid	CIF
sonnet-5.5	99.3%	3.30
haiku-5.5	98.3%	2.67
sonnet-4.5	79.9%	2.26
haiku-4.5	73.3%	3.11

CIF = canonical answer produced ÷ expected under uniform choice among valid answers.

H1 floor gone

Supported: 4 of 4 above 30%

H2 detection loss

Supported: E2 detects in 3.0% of draws

H3 canonical inflation

Supported: CIF 2.3-3.3 (all > 1)

H4 ranking distortion > 1.5x

NOT supported: max 1.43x

Local models (floor: 6.6-30.9% valid)

Kendall tau-b 0.62 ± 0.35 across 50,000 reference sets. Winner reverses in 17.56% of pairs, but 85.6% of that sits in pairs even the validator cannot separate. The validator finds qwen3 > llama3.1 and > mistral in every draw; E2 detects it in 5 of 50,000 (99.99% missed).

Above the floor

Uniform-reference winner reversal rises to 26.9%. Matching k=5 references detects 36-44% of real differences, k=25 detects 51-69%, a single reference only 2-5%.

Honest limits

One 24-task domain. Plan is an internal freeze. CLI sampling is provider-default. Metrics defined post hoc. Prevalence of single-reference scoring unaudited. Empirical study, not a proof; workshop-level.

Conclusion

What we built

A grounded, streaming OS tutor that explains first, then asks one graded Socratic question (hints, misconception probes, quiz, follow-up chips); and a research study of reference choice on 24 OS tasks with 8 models.

What we learned

Chatbot: defects and the prompt beat retrieval technique; measure the noise (±0.027). Research: one stored gold answer hides real model differences (detected in 0.01-3% of draws) and models over-produce the canonical answer 2.3-3.3x; a validator avoids both.

Limits and next

26 questions, 7B judge, no learner study. Research: one 24-task domain, internal preregistration, H4 failed, prevalence unaudited. Next: second domain with 65+ tasks, external registration, learner study.

Code, evidence and findings log

github.com/adhyan-jain/OS-Tutor-RAG (FINDINGS.md · paper/PAPER_FINAL.md = RCR paper)

References

[1] Lewis et al. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. NeurIPS 33. arXiv:2005.11401

[2] Karpukhin et al. (2020). Dense passage retrieval for open-domain question answering. EMNLP 2020. arXiv:2004.04906

[3] Gao et al. (2022). Precise zero-shot dense retrieval without relevance labels (HyDE). arXiv:2212.10496

[4] Es et al. (2023). RAGAS: Automated evaluation of retrieval augmented generation. arXiv:2309.15217

[5] Zheng et al. (2023). Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. arXiv:2306.05685

[6] Xiao et al. (2023). C-Pack: Packed Resources For General Chinese Embeddings (BGE). arXiv:2309.07597

[7] Wood, Bruner & Ross (1976). The role of tutoring in problem solving. J. Child Psychol. Psychiatry 17(2), 89-100

[8] Arpaci-Dusseau & Arpaci-Dusseau. Operating Systems: Three Easy Pieces (course reading)

[9] Casola et al. (2025). References Matter: reference set variation in summarization evaluation. INLG 2025. arXiv:2506.14335

[10] Jeong et al. (2026). Reporting Practice Matters: reference choice in chest X-ray report evaluation. arXiv:2609.19093

[11] Chen et al. (2026). HypoSpace: set-valued hypothesis generation under underdetermination. arXiv:2510.15614 (ICML 2026)

[12] OS-Tutor-RAG repository: github.com/adhyan-jain/OS-Tutor-RAG

arXiv identifiers [1]-[6] and [9]-[11] checked against arxiv.org; [7] and [8] are standard journal/textbook citations.